Regulation (ESMB) 2023/14 on Harmonised Indices of Consumer Prices

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Preamble

THE EUROPEAN STATISTICAL METHODS BOARD, having regard to its Statute, and in particular the provisions on common statistical standards, and after consulting the Committee of Member Statistical Authorities, has adopted this Regulation. The considerations set out below form part of the reasons for its adoption.

(a) Harmonised indices of consumer prices (HICPs) are the reference measure of consumer price inflation for monetary policy and for the assessment of price convergence among the participating countries. They must therefore be comparable across countries and over time.

(b) Comparability requires common rules on coverage, weighting, the index formula, the treatment of quality change, the index reference period and the timing of transmission. Differences between national consumer price indices (CPIs) and HICPs are legitimate where national indices serve other purposes, such as indexation of wages or pensions, provided that users are informed of those differences.

(c) The index reference period of the HICP must be changed from time to time so that the published index levels remain easy to interpret. Changing the index reference period does not change the rates of change of the index other than through rounding.

Chapter I General provisions

Article 1 Subject matter and scope

(1) This Regulation lays down the rules for the compilation, transmission and dissemination of harmonised indices of consumer prices by the Member Statistical Authorities (MSAs) participating in the European Statistical Methods Board.

(2) It applies to the all-items HICP, to all sub-indices at the level of the classification set out in Annex I, to the HICP at constant tax rates and to the special aggregates listed in Article 8.

Article 2 Definitions

For the purposes of this Regulation, the following definitions apply:

(a) "harmonised index of consumer prices (HICP)" means a Laspeyres-type price index measuring the change over time in the prices of goods and services acquired by households in the economic territory of the country, compiled in accordance with this Regulation;

(b) "household final monetary consumption expenditure (HFMCE)" means that part of final consumption expenditure incurred by households, irrespective of nationality or residence status, in monetary transactions on the economic territory of the country, on goods and services used for the direct satisfaction of individual needs or wants (the domestic concept);

(c) "index reference period" (also referred to in this Regulation as the "base year") means the period for which the index is set equal to 100;

(d) "weight reference period" means the period whose expenditure pattern is used to derive the weights;

(e) "price reference period" means the period with which the prices of the current period are compared; for the HICP it is December of the preceding year;

(f) "chain-linking" means joining two consecutive index segments, each with its own weights and price reference period, so that a continuous series is obtained;

(g) "quality adjustment" means the removal, from an observed price difference between two products, of the part that is due to a difference in quality, so that only pure price change is recorded in the index;

(h) "flash estimate" means an early estimate of the all-items HICP and of main aggregates, published before the end of the reference month or shortly after it.

Article 3 Coverage

(1) The HICP shall cover all goods and services included in HFMCE, classified according to the European Classification of Individual Consumption according to Purpose (ECOICOP) derived from COICOP 2018, as listed in Annex I.

(2) The following shall be excluded: owner-occupied housing services (imputed rents), games of chance, life insurance, interest payments and expenditure on narcotics.

(3) Prices shall be the prices actually paid by households, including taxes on products such as value added tax and excise duties, and net of subsidies on products. Reimbursements made by social security schemes shall be deducted where they reduce the price paid by the household.

(4) The geographical coverage shall be the whole economic territory of the country. Prices collected in a sample of localities shall be representative of the whole territory.

Chapter II Compilation

Article 4 Index formula and annual chain-linking

(1) The HICP shall be an annually chain-linked Laspeyres-type index. Within each calendar year t, the index shall compare the prices of the current month with the prices of December of year t-1 using fixed weights.

(2) Each annual segment shall be chain-linked to the previous one using December of year t-1 as the overlap month (the one-month overlap method). For an aggregate index I, the linked index for month m of year t is obtained as I(t,m) = I(t-1,Dec) x [segment index (t,m) / 100], where the segment index is expressed with December of t-1 = 100.

(3) As a consequence of chain-linking, sub-indices are not additive: an aggregate cannot be reproduced exactly by weighting the published sub-indices with the current weights over periods longer than one calendar year.

(4) Elementary aggregates shall be compiled with the ratio of geometric means (Jevons) or the ratio of arithmetic means (Dutot). The Carli formula (arithmetic mean of price relatives) shall not be used, because it is upward biased.

Article 5 Weights

(1) Weights shall reflect the HFMCE of the year t-1. Where data for t-1 are not yet final, the best available estimate shall be used and shall be reviewed against final data within two years.

(2) Weights derived from expenditure in the weight reference period shall be price-updated to December of year t-1 using the price changes of the corresponding sub-indices.

(3) Weights shall be expressed in parts per thousand of the all-items HICP and shall be published with the January index of each year.

Article 6 Index reference period (base year)

(1) The common index reference period for the HICP shall be 2015=100. All indices transmitted under this Regulation shall be expressed on that base until the change provided for in Article 12 takes effect.

(2) Indices compiled by an MSA on a different base, for national purposes, shall not be presented as HICPs. Where such indices are published alongside the HICP, the base year shall be stated next to each series, for example "CPI, 2021=100" and "HICP, 2015=100".

(3) An index expressed on one base can be converted to another base by dividing every value by the average index of the new base year computed on the old base, and multiplying by 100. Rebasing changes index levels but leaves rates of change unchanged except for rounding differences, as illustrated in Annex II.

(4) Index levels expressed on different bases shall not be compared, averaged or combined with each other. A difference in index levels caused only by a different base year is not a difference in price developments.

Article 7 Seasonal items and quality change

(1) Seasonal items, such as clothing, footwear and fresh fruit and vegetables, shall be included in every month in which they are available. During out-of-season months, prices shall be estimated so that the index reflects the price development of comparable in-season items.

(2) When a product in the sample is replaced by a product of different quality, the price difference shall be quality-adjusted using explicit methods (hedonic regression, option pricing, quantity adjustment) or, where those are not feasible, implicit methods such as bridged overlap or class-mean imputation.

(3) Automatic linking that treats the entire price difference between an old and a new product as a quality difference shall be avoided where it is known to bias the index.

Article 8 Constant-tax HICP and special aggregates

(1) MSAs shall compile an HICP at constant tax rates (HICP-CT), in which taxes on products are held at the rates of the price reference period. The difference between the HICP and the HICP-CT approximates the impact of tax changes on inflation, assuming full and immediate pass-through.

(2) MSAs shall compile at least the following special aggregates: energy; food including alcohol and tobacco; processed food; unprocessed food; non-energy industrial goods; services; all-items excluding energy and food ("core" inflation); administered prices.

Chapter III Transmission and dissemination

Article 9 Timeliness

(1) The flash estimate of the all-items HICP and of the main special aggregates shall be transmitted no later than the last working day of the reference month.

(2) The complete set of HICP indices shall be transmitted no later than 15 calendar days after the end of the reference month.

(3) Indices shall be transmitted with two decimal places.

Article 10 Rates of change and rounding

(1) Monthly and annual rates of change shall be computed from the indices rounded to two decimal places and published rounded to one decimal place.

(2) The annual rate of change for month m of year t is the percentage change between the index of month m of year t and the index of month m of year t-1. The annual average rate of change is the percentage change between the average index of year t and the average index of year t-1.

(3) Changes in rates of change shall be expressed in percentage points. A rise of the annual rate from 4.6 % to 5.1 % is an increase of 0.5 percentage points, not an increase of 0.5 %.

Article 11 Revisions

(1) The HICP shall not be revised as a routine matter. Revisions shall be made to correct errors, to incorporate improved weights under Article 5(1), or following changes to this Regulation.

(2) The flash estimate is superseded by the complete index; differences between the two are not revisions.

(3) Revisions shall be announced in advance where possible and documented in accordance with the ESMB Data Quality and Revisions Policy.

Article 12 Change of the index reference period

(1) The Board may decide a future change of the index reference period by an amending regulation, which shall fix the new reference period and the first index released on it, and shall require complete back series on the new base. Until such a change takes effect, 2015=100 applies.

(2) MSAs shall not transmit HICPs on any base other than 2015=100.

Article 13 Entry into force

This Regulation shall enter into force on the twentieth day following its publication in the ESMB Official Bulletin. It shall be binding in its entirety on all participating MSAs.

Annex I ECOICOP divisions

Division	Title
01	Food and non-alcoholic beverages
02	Alcoholic beverages, tobacco and narcotics
03	Clothing and footwear
04	Housing, water, electricity, gas and other fuels
05	Furnishings, household equipment and routine household maintenance
06	Health
07	Transport
08	Information and communication
09	Recreation, sport and culture
10	Education services
11	Restaurants and accommodation services
12	Insurance and financial services
13	Personal care, social protection and miscellaneous goods and services

Narcotics in division 02 are excluded from coverage under Article 3(2).

Annex II Illustrative example of a change of base year

The table shows an index expressed on 2015=100 and the same index converted to 2021=100. The average of the twelve monthly indices of 2021 on the 2015=100 base is assumed to be 112.40. Each value on the new base equals the old value divided by 112.40 and multiplied by 100. The figures are illustrative.

Month	Index, 2015=100	Index, 2021=100	Annual rate, %
January 2023	136.92	121.81	13.2
January 2024	146.83	130.63	7.2
January 2025	154.02	137.03	4.9

The annual rates are identical on both bases: for January 2024, 146.83 / 136.92 = 1.0724 and 130.63 / 121.81 = 1.0724, an increase of 7.2 %. Only the levels differ. A user who places the two index columns side by side without the base year will see an apparent discrepancy of 15 to 17 index points that has no economic meaning.